

Web Frameworks for DS/AI Students

Web Frameworks for DS and AI Students:

- AI/ML models are not useful if they just run on Jupyter Notebook.
- We need a web framework to:
 - Deploy models
 - Create simple dashboards
- Build APIs so other applications can use predictions.

◆ Django (Full-Stack Framework)

- Best for: Building complete web apps.
- Comes with ORM, authentication, admin panel, database support.
- Used by companies like *Instagram, Pinterest*.
- Great when:
 - You want end-to-end apps (front-end + backend).
 - You want to deploy ML models on the website.

❖ Fastapi (API Framework)

- Best for: Deploying ML models quickly
- Lightweight, faster than Django (because async).
- Easy to build REST APIs → great for serving AI/ML models.
- Doesn't force front-end, but can connect with React/Angular if needed.
- Used in industry for ML model deployment.

- **Django:**

Django Architecture

Django follows the MTV architecture:

1.**Model** → Defines your database structure (*tables*).

Example: models.py

2.**Template** → Defines the front-end (*HTML, CSS*).

Example: templates/ folder

3.**View** → Connects Model and Template. Handles logic & data flow.

Example: views.py